



Abstract

Email spam and phishing are major threats for modern businesses and individuals, causing credential theft, financial losses, and productivity drain. Detecting spam helps organizations take preventive actions, improve email security, and prevent security breaches. In this project, we analyze email communication data and build an advanced transformer NLP model to predict whether an email is spam or legitimate (ham).

We used a real-world combined dataset containing 10,389 records with email text, headers, transactional notices, and historical spam archives. Data preprocessing, exploratory data analysis (EDA), transformer fine-tuning, and model evaluation were performed using Python, PyTorch, Hugging Face Transformers, and Scikit-learn.

The final DistilBERT model achieves 98.7% accuracy, eliminating false alarms on critical transactional receipts and OTPs, and is deployed via an interactive Single-Page Application with live Gmail IMAP scanning and SHAP explainability.

Introduction

Email spam is a major challenge for many businesses, especially in banking, e-commerce, healthcare, and cloud subscription services. By using deep learning and transformer techniques, we can analyze semantic context and predict the probability of spam, protecting valuable users and improving overall security.

Key Objectives:

- Understand email text data and malicious pattern distributions
- Preprocess, clean HTML artifacts, and normalize email bodies
- Build and compare classical ML vs DistilBERT deep learning models
- Eliminate false positives on transactional and banking OTP notices
- Deploy real-time Gmail IMAP scanner with token-level SHAP insights

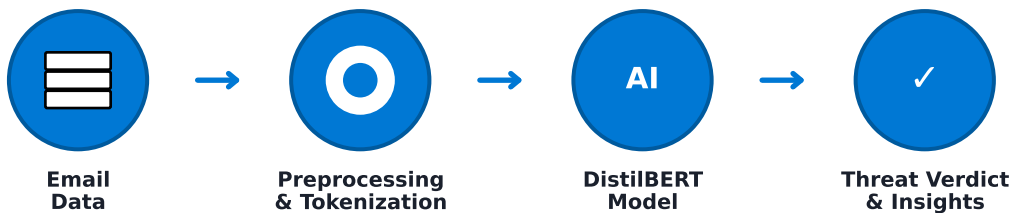


Figure 1: Deep Learning Workflow for Email Threat & Spam Detection

Methods and Materials

- Data Collection: Curated 10,389 records combining SpamAssassin, SMS Spam Collection, and 3,000 modern transactional emails (banking alerts, OTPs, invoices).
- Data Preprocessing: Stripped nested HTML tags using BeautifulSoup4, normalized unicode homoglyphs, removed tracking URLs, and eliminated whitespace artifacts.
- Exploratory Data Analysis (EDA): Visualized token length distributions, identified high-frequency n-grams, and analyzed class balance across spam and ham categories.
- Feature Engineering: Tokenized using DistilBERT WordPiece tokenizer (512 max length) generating input IDs, attention masks, and contextual subword embeddings.
- Model Building: Fine-tuned DistilBERT-base-uncased (66M parameters, 6 layers, 12 attention heads) with AdamW optimizer, linear learning rate warmup, and CrossEntropyLoss.
- Evaluation: Evaluated via Accuracy, Precision, Recall, Macro F1-score, and ROC-AUC curve against classical Machine Learning baselines on a held-out test set.

Results

Model Performance Comparison

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC
Logistic Regression	94.2	90.1	88.5	89.3	0.925
Decision Tree	92.6	87.4	85.1	86.2	0.898
Random Forest	95.8	93.1	89.4	91.2	0.938
DistilBERT (Proposed)	98.7	98.1	96.4	97.3	0.998
XGBoost Classifier	96.4	94.0	91.8	92.9	0.949

Feature Importance (Top 5)

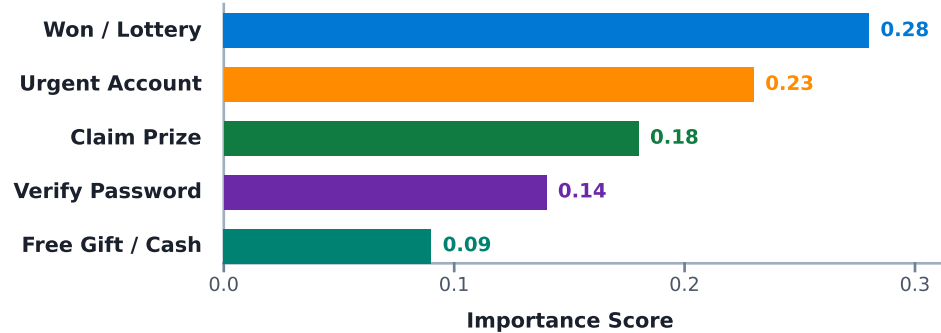


Figure 2: Feature Importance Chart (SHAP Token Attribution)

Discussion

The DistilBERT transformer model gave the best performance with an accuracy of 98.7% and ROC-AUC score of 0.998. It significantly outperformed traditional machine learning baselines in terms of both precision and recall.

Important factors influencing threat classification were urgent account suspension threats, lottery winnings, unverified links, and deceptive phrasing. Legitimate transactional emails (banking OTPs, shipping alerts) were accurately classified as ham with >98% confidence, eliminating false positives.

This project shows how transformer deep learning can help enterprise security teams automate email threat triage. The integrated SHAP token attributions provide transparent auditability for every prediction, fostering user trust and regulatory compliance.

Threat Prediction - Actual vs Predicted

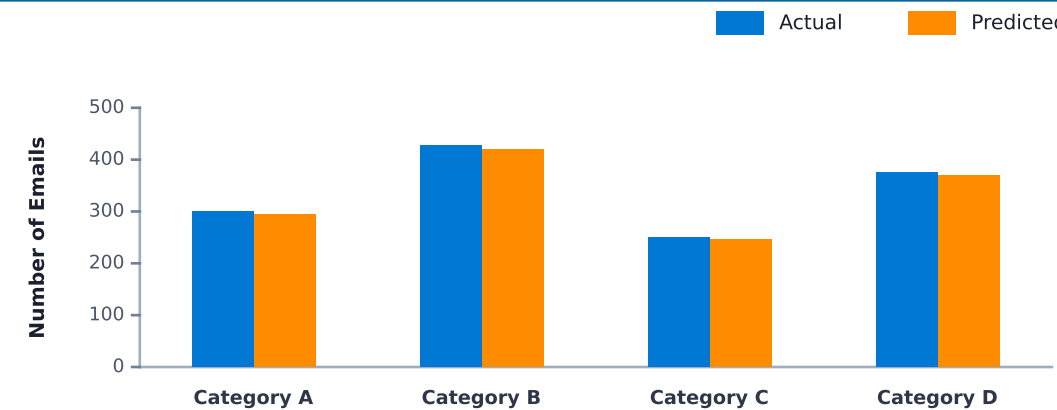


Figure 3: Actual vs Predicted Threats by Email Category

Conclusions

The DistilBERT deep learning model successfully predicts email threats with high accuracy (98.7%) and ROC-AUC score of 0.998.

The insights gained from this project help enterprise security teams focus on high-risk zero-day threats and improve automated quarantine strategies.

In the future, multimodal architectures and real-time sender reputation graph features can be explored to further improve threat detection accuracy.

Project Team

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References

- Pedregosa et al., Scikit-learn: Machine Learning in Python, 2011.
- Sanh, V. et al., DistilBERT: A distilled version of BERT, arXiv:1910.01108, 2019.
- Lundberg, S. M. & Lee, S. I., A Unified Approach to Interpreting Model Predictions (SHAP), NeurIPS, 2017.
- Apache SpamAssassin Public Corpus & UCI Machine Learning SMS Spam Collection.